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Experiment No 7

06/10

Title :- Data Cleaning

Aim :- To understand and apply techniques for handling missing data and finding outliers using Python with Numpy and Pandas, focusing on practical applications in data science and machine learning.

Theory :-

1) What is meant by missing data? Why is it important to handle missing value in a dataset?

=> Missing data means that no value is stored for a variable in an observation, and pandas shows it as NaN. It arises from skipped survey questions, sensor failures, entry mistakes or failed data merges. Handling it is important because most machine learning algorithm cannot accept NaN, and calculations such as mean or sum before inaccurate. Ignoring missing values also reduces the usable data and can introduce bias in the results.

Code :-

```
import pandas as pd
import numpy as np
df = pd.DataFrame({"Name": ["A", "B", "C", "D"],
                   "Age": [21, np.nan, 23, 22],
                   "Marks": [78, 85, np.nan, np.nan]})
print(df.isnull().sum())
print(df.dropna())
```

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output :

```
Name      0
Age       1
Marks     2
dtype: int64
   Name   Age   Marks
0     A  21.0    78.0
```

Q2 Explain the difference between mean imputation and median imputation. In which case is median preffd preferred?

⇒

Point	Mean Imputation	Median Imputation
value used	Average of of all available value	Middle value of the sorted available values.
Effect of outliers	Pulled strongly towards extreme values	Not affected by extreme values.
suitable for	Symmetric data without outliers	skewed data or data with outliers.
syntax	fillna(col.mean())	fillna(col.median())

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Median is preferred when the column is skewed or has outliers, for example salary because a few very large value raise the mean and the filled values would then be unrealistically high.

code:

```
import pandas as pd
import numpy as np

salary = pd.Series([30,32,35,np.nan,38,400])
print(salary.fillna(salary.mean()))
print(salary.fillna(salary.median()))
```

output:

```
0     30.0
1     32.0
2     35.0
3    107.0
4     38.0
5    400.0
dtype: float64
0     30.0
1     32.0
2     35.0
3     35.0
4     38.0
5    400.0
dtype: float64
```

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Q3 What is an outlier? How do outliers affect the performance of Machine learning models?

=> An outlier is a data point that lies far away from the other observation in a dataset. It may come from a measurement error, a data entry mistake or a genuine rare event. Outliers pull the mean and increase the standard deviation, so the statistics no longer describe the normal data. Models such as linear regression and distance based methods are sensitive to them because the fitted line or the cluster centres shift and the accuracy on normal data falls.

code:

```
import numpy as np
data = np.array([52, 55, 53, 58, 54, 56, 300])
print("Mean with outlier", round(np.mean(data), 2))
print("Mean without outlier", round(np.mean(data[:6]), 2))
print("Std with outlier", round(np.std(data), 2))
print("std without outlier", round(np.std(data[:6]), 2))
```

output:

```
Mean with outlier 89.71
Mean without Outlier 54.69
Std with outlier 85.87
Std without outlier 1.97
```

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Q4 Discuss how improper handling of missing values and outliers can bias ~~model~~ machine learning models?

⇒ If rows with missing values are simply dropped, a large part of the data may be lost, and if the values are not missing at random the remaining rows no longer represent the population, so the model learns a biased pattern. Filling every gap with the mean reduces the variance of the column ~~bias~~ and weakens its relationship with other variables. Outliers that are left in the data pull the learned parameters towards themselves, while outliers removed without checking may delete genuine extreme cases. The model then performs well on the training data but gives poor predictions on new data.

code:-

```
import pandas as pd
import numpy as np
x = pd.Series([10, 20, 30, 40, 50, np.nan, np.nan, np.nan])
print("std before filling", round(x.std(), 2))
print("std after mean filling", round(x.fillna(x.mean()).std(), 2))
```

output:- std before filling 15.81
std after mean filling 11.95

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Q5 Suggest an approach to handle missing categorical data.

→ Missing categorical values cannot be filled with mean or median. A common approach is mode imputation, where the most frequent category replaces the gaps. Another approach is to create a separate category such as unknown, which also keeps the information that the value was missing. If only a few rows are affected, they can be dropped.

code:

```
import pandas as pd
import numpy as np

df = pd.DataFrame({"City": ["Mumbai", "Pune",
                            np.nan, "Mumbai",
                            np.nan]})
df["Mode"] = df["City"].fillna(df["City"].mode()[0])
df["Unknown"] = df["City"].fillna("Unknown")
```

output:

	City	Mode	Unknown
0	Mumbai	Mumbai	Mumbai
1	Pune	Pune	Pune
2	NaN	Mumbai	Unknown
3	Mumbai	Mumbai	Mumbai
4	NaN	Mumbai	Unknown

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Q 6 Can outliers ever be useful? Give an example from real-world data.

=> Yes an outlier is not always an Error, and sometimes it is the most important observation. In banking, a customer who normally spends a few thousand rupees and suddenly makes a transaction of several lakhs is given an outlier, and this is exactly the pattern a fraud detection system looks for. In the same way, an unusually high network traffic reading can point to an intrusion, and a very high body temperature shows a patient who needs attention.

Code :-

```
import pandas as pd
txn = pd.DataFrame({"Amount": [2500, 3200,
                               1800, 2900,
                               950000, 3100]})
print(txn[txn["Amount"] > 100000])
```

output :-

```
   Amount
4  950000
```

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Q7 What are the advantages and limitations of using z-scores for outlier detection?

=> The z score tells how many standard deviations a value lies from the mean. It is calculated as the value minus the mean, divided by the standard deviation, and a value with an absolute z score above 3 is usually treated as an outlier

Advantages	Dis Limitations
simple and quick to calculate using numpy or pandas	Mean and standard deviation are themselves affected by outliers, so extreme values can hide themselves.
Gives a standard scale, so different columns can be compared	Works well only when the data is roughly normally distributed
Threshold of 3 is easy to apply and explain	Not reliable for small datasets.

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code :

```
import numpy as np

data = np.array ([50,52,49,51,53,48,50,
                  52,51,49,50,53,47,
                  52,51,50,49,52,51,120])
z = (data - np.mean(data)) / np.std(data)
print("z score of last value", round(z[19],2))
print("Outliers", data[np.abs(z)>3])
```

output :-

```
z score of last value 4.34
outliers [120]
```

conclusions :-

Missing values were identified and handled using Numby and pandas by dropping them and by mean, median and mode imputation. outliers were detected using the z score method, and their effect on the mean and standard deviation was observed. choosing a suitable method for each column keeps the data reliable for training machine learning models.

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